

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 2ND SEMESTER FIRST INTERNAL EXAMINATIONS,

APRIL 2024

23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++(IT-A/IT-B /ECE-B)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the basic concepts of Object-Oriented Programming.
CO2	Describe the object-oriented paradigm with concepts of streams, classes, functions, data and objects.
CO3	Implement object-oriented programming constructs like encapsulation, inheritance and polymorphism.
CO4	Understand dynamic memory management techniques using pointers, constructors, destructors, etc.
CO5	Identify classes including data, methods and the relationship among the classes from a given problem statement and solve the problem using object oriented constructs in C++.

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No	Questions	Marks	BL	CO	PO
I (a)	What is procedure oriented programming?What are its main characteristics?	2	L1	1	1,12
(b)	What is the need of inline function?Write a C++ program to find the area of a circle using inline function.	2	L3	1	1,12
(c)	What are the different ways to define member functions of a class? Explain with an example.	2	L2	2	1,2,12
(d)	Explain destructors in C++ with an example..	2	L2	2	1,2,12
(e)	Explain the use of 'this' pointer.	2	L2	2	1,2,12

PART B

(Answer the following questions
choosing either the first OR second option)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Discuss the features and advantages of Object Oriented Programming.	7	L2	1	1,12

(1 x 10= 10 Marks)

(b)	Write a C++ program to find the sum of digits of a number.	3	L3	1	1,12
OR					
III (a)	Compare and contrast procedure oriented programming and object oriented programming approaches.	6	L2	1	1,12
(b)	Differentiate between formatted and unformatted I/O operations in C++ with suitable examples.	4	L2	1	1,12
(1 x 10= 10 Marks)					
IV (a)	What is a constructor? Discuss different types of constructor with the help of a C++ program.	6	L2	2	1,2,12
(b)	What is the significance of static data and member functions in C++.	4	L2	2	1,2,12
OR					
V (a)	Explain in detail about friend function. How do they differ from member functions, and in what scenarios are they useful?	7	L2	2	1,2,12
(b)	What is constructor overloading? Explain with an example.	3	L2	2	1,2,12

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
4	86	10	-	-	-